

PREDICTION TASK  What is the type of task? Which entity are predictions made on? What are the possible outcomes to predict? When are outcomes observed? Type of task: - Find similar photos and put them into categories (like selfie or screenshot) What do we make predictions on? - Each photo on the user's phone What are we trying to predict? - Are two or more photos similar or the same? When are predictions made? - Right after the user starts a scan	DECISIONS  How are predictions turned into actionable recommendations or decisions for the end-user? (Mention parameters of the process/application for this.) The app compares photos using AI (ResNet-50 + cosine similarity) It shows similar photos in groups The user can: - Keep the best one - Delete the rest - Swipe to choose Settings include: - How similar do photos need to be - Which types to clean up (like only screenshots)	VALUE PROPOSITION  Who is the end beneficiary, and what specific pain points are addressed? How will the ML solution integrate with their workflow, and through which user interfaces? Who benefits from this app? - People who want to free up storage and care about privacy What problems does it solve? - Too many photos, hard to clean manually - Storage runs out quickly - Cloud apps may risk privacy How does it fit into daily life? - Everything runs on the device - Simple UI with swipe actions - The user always controls what is deleted User interface: - iOS app with photo previews and category filters	DATA COLLECTION  How is the initial set of entities and outcomes sourced (e.g., database extracts, API pulls, manual labeling)? What strategies are in place to update data continuously while controlling cost and maintaining freshness? Where do we get the data? - From the phone's photo library (using Apple's Photos API) - We use photo info like size, date, and type to help with prediction How do we keep data up to date? - Automatically scan new photos - Users can tap "rescan" anytime - No cloud or API costs, all local	DATA SOURCES  Where can we get data on entities and observed outcomes? (Mention internal and external database tables or API methods.) Internal: - Photos API (photo info) - CoreML model (ResNet-50) for photo analysis - Storage info from the device External: - None. No internet or cloud data used
IMPACT SIMULATION  What are the cost/gain values for (in)correct decisions? Which data is used to simulate pre-deployment impact? What are the criteria for deployment? Are there fairness constraints? What happens with correct or wrong decisions? - Correct: Frees up space → good result - Wrong: Deletes important photo → bad result How do we test this before launch? - Use test photos and fake duplicates - Try deleting, with the "undo" option available Requirements for launch: - ResNet50 with precision over 80% - Fast on iPhone A12 or newer - No crashes when scanning lots of	MAKING PREDICTIONS  Are predictions made in batch or in real time? How frequently? How much time is available for this (including featurization and decisions)? Which computational resources are used? When are predictions made? - In batch (all at once) during each scan How often? - When user opens app or adds many new photos How long can it take? - Under 120 seconds for ~2,000 photos What resources are used? - iPhone's GPU / Neural Engine - Apple's CoreML + Swift concurrency		BUILDING MODELS  How many models are needed in production? When should they be updated? How much time is available for this (including featurization and analysis)? Which computation resources are used? How many models do we use? - Just one: ResNet-50 (CoreML) When do we update it? - Only when the app is updated Training time needed? - No training on the device. All done before release Tools used: - CoreML converter + Xcode	FEATURES  What representations are used for entities at prediction time? What aggregations or transformations are applied to raw data sources? What features are used to make predictions? - AI image embeddings What data transformations are done? - Compare embeddings using cosine similarity - Group similar images - Use file names (e.g. "screenshot") to help with classification

photos

Fairness:

- Never delete without asking
- Add undo or "recently deleted" feature

MONITORING

Which metrics and KPIs are used to track the ML solution's impact once deployed, both for end-users and for the business?
How often should they be reviewed?

What metrics do we track?

User experience:

- How much storage is saved
- Accuracy (based on undo rate)
- Time from scan to cleanup

Business:

- Active users (daily/monthly)
- Ad revenue (Google Ads)
- App stability (no crashes)

How often do we check?

- Weekly for user activity
- Monthly for model and app updates



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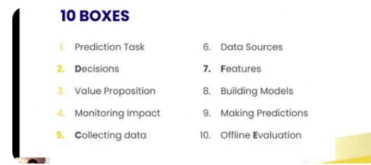
[FREE] ONLINE COURSE

Introduction to the Machine Learning Canvas

Get started with the MLC in this short course taught by its author.



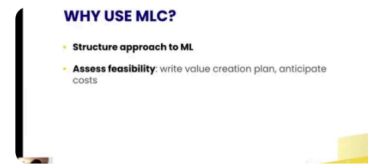
Start Here



[Video] Overview of the 10 boxes that make up the MLC



3:31



[Video] Why use the MLC?



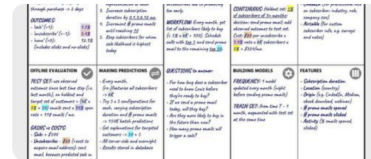
2:34



[Video] Structure of the MLC



2:49



Get your AI-generated canvas



Prepare for implementation



Start now at ownml.co/intro